MMDU MULLANA OFFICIAL SYLLABUS • BCSE-012 • UNIT 1

Unit 1: Internet & World Wide Web Architecture (Complete Theory & Schematics)

Subject: Fundamentals of Web Technologies (BCSE-012) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

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1. Technical Definitions & 12-Point Internet vs. WWW Analysis

1. Fundamental Principles: The Internet vs. The World Wide Web

Understanding the structural boundary between the Internet transport fabric and the World Wide Web hypermedia application is the cornerstone of web engineering.

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous Technical Definition: The Internet

> A globally coordinated, packet-switched network of interconnected computing hardware and communications infrastructure governed by the **TCP/IP protocol suite**. It provides transparent physical transport pipelines for routing raw binary datagrams across subsea fiber cables, satellite uplinks, and enterprise routers regardless of device hardware vendor.

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous Technical Definition: The World Wide Web (WWW)

> An abstract cyberspace application and distributed hypermedia information retrieval system operating atop the physical Internet infrastructure. It consists of interlinked documents and multimedia resources addressed globally via **Uniform Resource Locators (URLs)** and transferred using **HTTP/HTTPS**.

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2. Comprehensive 12-Point Comparative Matrix

In university examinations, differentiating the Internet from the World Wide Web carries 6 to 10 marks:

#	Dimension	The Internet	The World Wide Web (WWW)
1	**Fundamental Nature**	Vast physical & logical network of interconnected hardware, cables, and routers.	Cyberspace collection of hyperlinked multimedia documents and online software resources.
2	**Year of Origin**	**1969** (Launch of ARPANET by US Department of Defense DARPA).	**1989** (Invented by Sir Tim Berners-Lee at CERN, Switzerland).
3	**OSI Layer**	Operates across lower layers: Physical, Data Link, Network, and Transport.	Operates strictly at Layer 7: the **Application Layer**.
4	**Governing Protocols**	Driven by IP \frac{IPv4}{IPv6}, TCP, UDP, BGP, ICMP, ARP, RIP.	Driven by HTTP, HTTPS, WebSockets, WebRTC.
5	**Core Purpose**	Transports raw binary packets between endpoints reliably.	Structures, formats, and navigates multimedia hypermedia.
6	**Dependency Relation**	**Completely independent.** It existed 20 years before the Web and functions without it.	**Completely dependent on the Internet.** The Web cannot exist without the Internet backbone.
7	**Addressing Scheme**	Hardware MAC addresses (48-bit) and logical IP addresses (32/128 bit).	Uniform Resource Locators (URLs) and URIs (e.g. `https://mmumullana.org`).
8	**Services Encompassed**	Encompasses the Web, Email (SMTP/IMAP), File Transfer (FTP), Shells (SSH), VoIP.	Encompasses websites, web applications, REST APIs, and hypermedia DOM trees.
9	**Client Software**	Operating system networking stack, terminal clients, socket utilities.	Web Browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).
10	**Physical Hardware**	Submarine fiber cables, repeaters, cellular towers, ISP core routers.	Web hosting server machines, disk arrays, HTTP daemons, database clusters.
11	**Central Concept**	Packet switching, packet headers, hop-by-hop routing, congestion control.	Hypertext, hyperlinks, DOM trees, CSS cascades, rendering engines.
12	**Governing Entities**	IETF (Internet Engineering Task Force), ICANN, IANA, ITU.	W3C (World Wide Web Consortium), WHATWG.

📘 **Level 1 (Easy): Categorizing Internet Services vs Web Applications**

Problem Statement:
Categorize each of the following technologies as belonging strictly to the **Internet Infrastructure Layer** or the **World Wide Web (WWW) Application Layer**:

1. Domain Name System (DNS) server operating on UDP port 53.
2. A single-page interactive student portal loaded over HTTPS.
3. An SSH terminal shell session connecting to a remote Linux server on port 22.
4. An HTML5 video element streaming via native browser controls.

Step 1 🚀: Analyze DNS (Port 53)

- Operates at the transport/network abstraction level to translate domain names into IP addresses.
- **Classification: Internet Infrastructure Layer**.

Step 2 ⚡: Analyze HTTPS Student Portal

- Involves browser DOM rendering, HTML/CSS assets, and HTTP application transactions.
- **Classification: World Wide Web Application Layer**.

Step 3 🎯: Analyze SSH Terminal (Port 22)

- Runs directly over TCP transport independently of web browsers and HTTP.
- **Classification: Internet Service (Non-Web)**.

Step 4 🏁: Analyze HTML5 Video Tag

- Embedded in HTML markup and parsed by the browser layout engine.
- **Classification: World Wide Web Application Layer**.

🎯 1: Internet Infrastructure | 2: World Wide Web | 3: Internet Service (Non-Web) | 4: World Wide Web.

2. Historical Evolution: ARPANET to Web 3.0 & Application Domains

1. Technological Milestones in Global Web History

The evolution of modern web computing spans five key milestones:

1. **1969 (ARPANET):** Funded by the US Defense Advanced Research Projects Agency (DARPA). It connected four research university nodes (UCLA, Stanford, UC Santa Barbara, Utah) via Interface Message Processors (IMPs) using

packet switching.

2. **January 1, 1983 ("Flag Day"):** ARPANET completed its mandatory migration from NCP to the **TCP/IP protocol suite**, marking the birth of the modern Internet.

3. **1989–1990 (Tim Berners-Lee at CERN):** Sir Tim Berners-Lee authored the proposal for an information management system using hyperlinked documents. He authored the first HTTP daemon (`CERN httpd`), the first graphical browser (`WorldWideWeb`), and the initial HTML specification.

4. **1993 (NCSA Mosaic):** Developed by Marc Andreessen at the University of Illinois Urbana-Champaign; introduced inline graphic rendering alongside text, accelerating consumer internet adoption.

5. **1994 (W3C Established):** Tim Berners-Lee founded the **World Wide Web Consortium (W3C)** at MIT to create vendor-neutral web standards.

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2. The Three Generations of the Web

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[ Web 1.0 (1991–2004) ]          [ Web 2.0 (2004–2016) ]          [ Web 3.0 (2016–Present) ]
     "The Read-Only Web"              "The Read-Write Web"             "The Semantic & AI Web"
  • Static HTML files              • User-generated content         • Machine-readable semantics
  • Table-based layouts            • AJAX asynchronous calls        • Decentralized storage (IPFS)
  • Centralized webmasters         • Social media & blogs           • AI search & Progressive Web Apps
  • Zero user interactivity        • Responsive mobile design       • Smart agents & WebAssembly
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3. Application Domains of the Internet

1. **Electronic Commerce (E-Commerce):** B2B, B2C, and C2C marketplaces with automated inventory management, UPI/credit-card payment gateways, and real-time shipment telemetry.

2. **Education & Remote Learning:** Virtual interactive classrooms, Massive Open Online Courses (MOOCs), video streaming of university lectures, and automated examination platforms.

3. **Telecommunications & Collaboration:** Voice over IP (VoIP), high-definition video conferencing, instant messaging, and distributed version control (Git/GitHub).

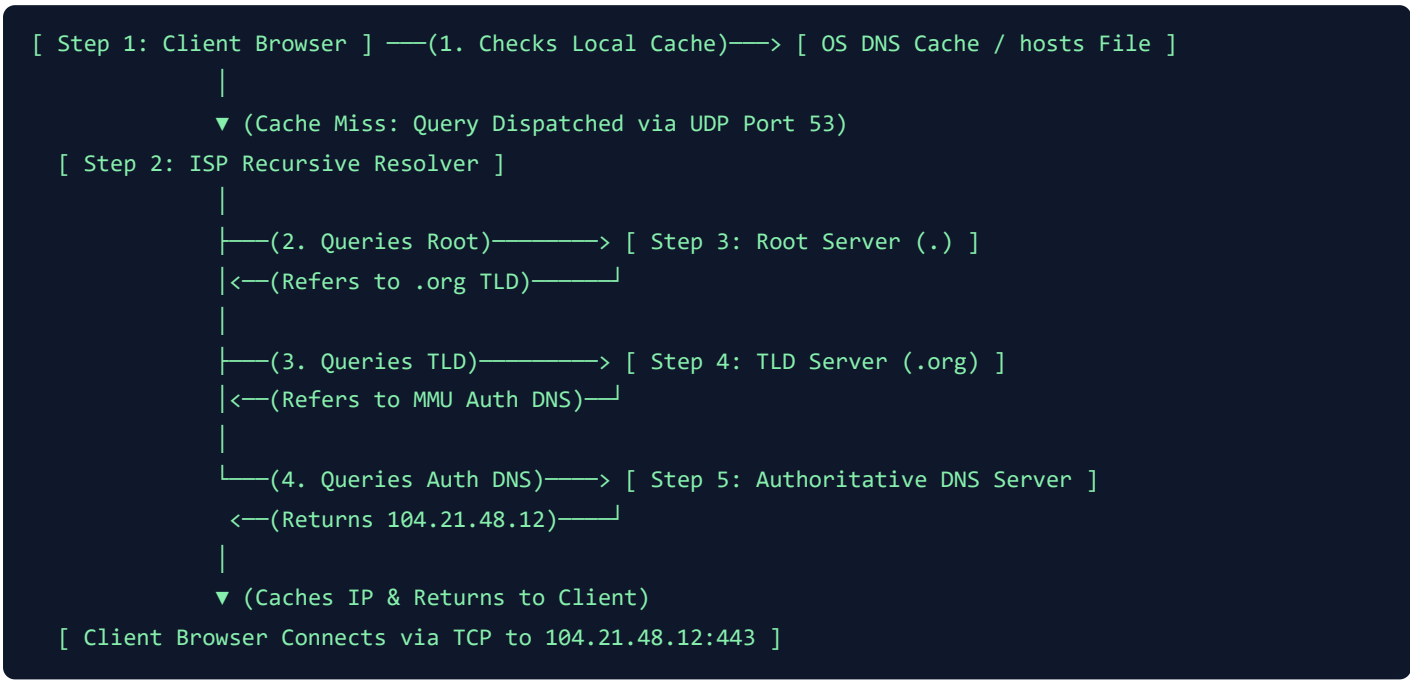
4. **Cloud Computing & Content Delivery:** Infrastructure as a Service (IaaS), Software as a Service (SaaS), automated container orchestration (Kubernetes), and global edge CDNs.

5. **Internet of Things (IoT) & Smart Cities:** Industrial telemetry, automated electrical grid monitoring, environmental sensing, and connected vehicles.

3. DNS Resolution Architecture & Web Client-Server Subsystems

1. Requirements for Working with Internet & WWW: The 5-Tier DNS Resolution Hierarchy

When a user enters `https://mmumullana.org` into a browser, the hostname must be resolved to a 32-bit IPv4 or 128-bit IPv6 address before TCP connection setup can occur:



> 📌 **CRITICAL EXAM DERIVATION:**
> **Key Architectural Distinction:**

> - **Recursive Resolver:** The server that performs the legwork of querying other nameservers on behalf of the client.

> - **Authoritative DNS:** The final authoritative source holding the actual DNS zone record (`A` , `AAAA` , `CNAME` , `MX`) mapped to that domain.

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2. Internal Subsystems of a Modern Web Browser

A web browser is a multi-process software application composed of distinct subsystems:



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3. Hypertext Transfer Protocol (HTTP) & The 3-Way Handshake

HTTP is an Application Layer, stateless, request-response communication protocol:

> $$\text{Client Request[Method + URI + Headers + Body]} \textit{iff} \text{Server Response[Status Code + Headers + Payload]}$$

Standard HTTP Methods:

- `GET` **:** Requests representation of specified resource without server side-effects. Parameters appended to URL query string.
- `POST` **:** Submits data to be processed (e.g. form submissions, file uploads). Data enclosed inside request payload body.

- `PUT` : Replaces target resource completely with uploaded payload.
- `DELETE` : Requests removal of specified resource from server storage.

HTTP Response Status Classifications:

- **1xx (Informational):** `100 Continue` , `101 Switching Protocols` .
- **2xx (Success):** `200 OK` , `201 Created` , `204 No Content` .
- **3xx (Redirection):** `301 Moved Permanently` , `304 Not Modified` (browser cache valid).
- **4xx (Client Error):** `400 Bad Request` , `401 Unauthorized` , `403 Forbidden` , `404 Not Found` .
- **5xx (Server Error):** `500 Internal Server Error` , `502 Bad Gateway` , `503 Service Unavailable` .

4. Web Searching Architectures & Web-Casting Techniques

1. Search Engine Subsystems & Inverted Indexing

Modern web search engines process billions of documents via a 4-phase pipeline:

```
[ Phase 1: Web Crawlers (Spiders / Bots) ]
  • Discovers URLs by traversing hyperlinks across the web graph.
  • Respects robots.txt directives and crawls recursively.
                    |
                    ▼
  [ Phase 2: Document Parser & Tokenizer ]
  • Strips HTML markup tags and extracts raw text.
  • Lowercases tokens, eliminates non-informative stop words ("the", "is", "at").
  • Normalizes words to linguistic stems via Porter Stemming.
                    |
                    ▼
  [ Phase 3: Inverted Index Generation ]
  • Constructs a dictionary of all distinct words mapped to Document Postings Lists:
      Term "html"    ——> [ Doc 101, Doc 104, Doc 209 ]
      Term "css"     ——> [ Doc 104, Doc 312 ]
      Term "mullana" ——> [ Doc 101, Doc 418 ]
                    |
                    ▼
  [ Phase 4: Query Ranking Engine (PageRank / AI) ]
  • Evaluates query terms using TF-IDF (Term Frequency - Inverse Document Frequency).
  • Computes PageRank authority score based on incoming link eigenvector centrality.
  • Returns top ranked results to user in under 100 milliseconds.
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2. Web-Casting Techniques: Push vs. Pull & Unicast vs. Multicast

- **Pull Technology (Traditional Client-Driven):** The client sends an explicit HTTP request to the server, and the server passively responds with data. If no request is made, no data is delivered.

- **Push Technology (Server-Driven Event Streams):** The server proactively delivers fresh information to the client as soon as new data becomes available without waiting for a new request (e.g. WebSockets, Server-Sent Events, RSS feeds).

- **Unicast Transmission:** A dedicated one-to-one point-to-point transmission stream between the server and a single client. N simultaneous viewers require N times the server bandwidth.

- **Multicast Transmission:** A one-to-many distribution mechanism where the source transmits a single packet stream, and intermediate network routers replicate packets at junction branches to feed multiple simultaneous viewers.

5. University Solved Exam Questions (With Schematics)

University Examination Solved Questions (Unit I)

The following questions feature diagrams in **both the question statement and the step-by-step solution**:

📘 Section A (1 Mark): Root Name Server Purpose in DNS

University Question (1 Mark):
State the primary purpose of the Root Name Server in the DNS resolution hierarchy shown in Figure Q1.

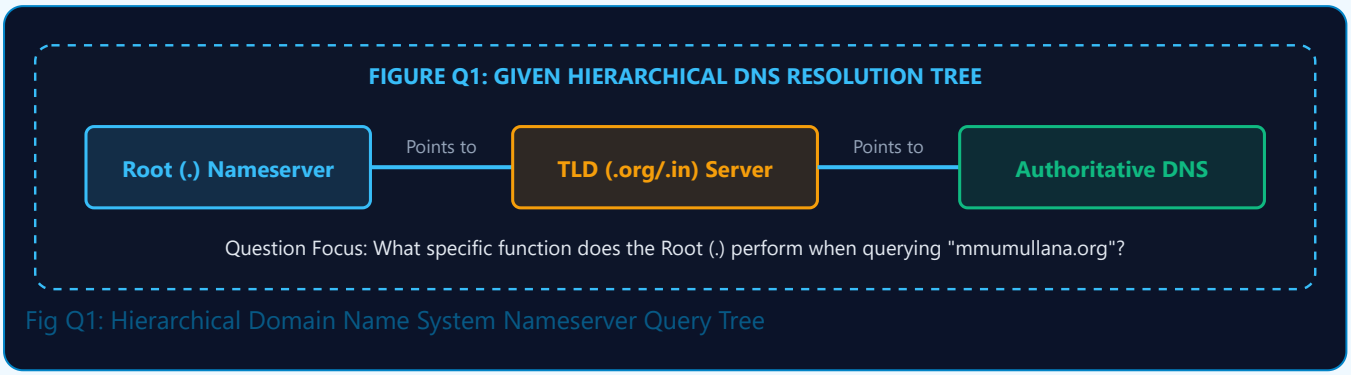


Fig Q1: Hierarchical Domain Name System Nameserver Query Tree

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Step-by-Step Solution:
The **Root Name Server (`.`)** stands at the apex of the global DNS namespace. It does not contain end-user host IPs; instead, it parses the Top-Level Domain suffix (e.g. `.org`) and returns the IP address of the designated **TLD Nameserver** to the recursive resolver.

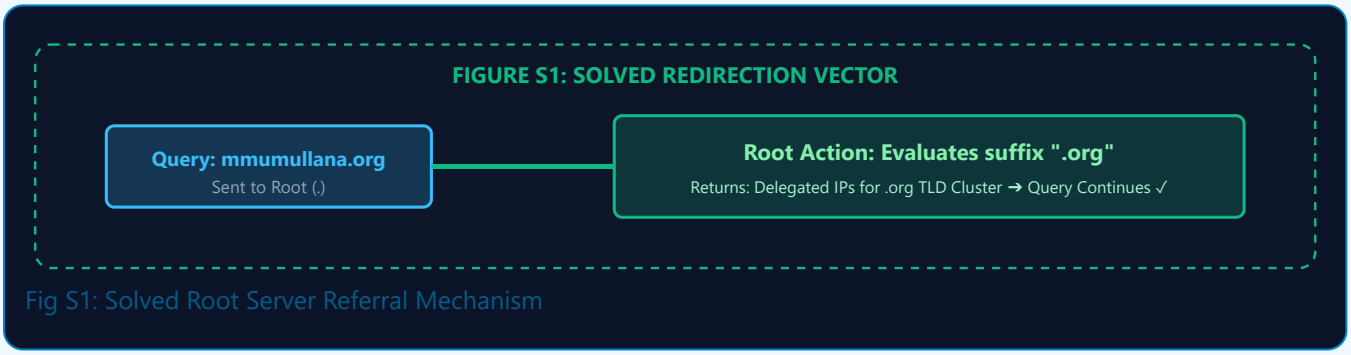


Fig S1: Solved Root Server Referral Mechanism

🎯 Root Name Server reads domain TLD (`.org`) and refers resolver to TLD Authoritative Cluster.

📘 Section B (2 Marks): TCP 3-Way Handshake Timing Sequence

University Question (2 Marks):
Illustrate the 3-way handshake connection sequence between a client browser and an Apache web server.

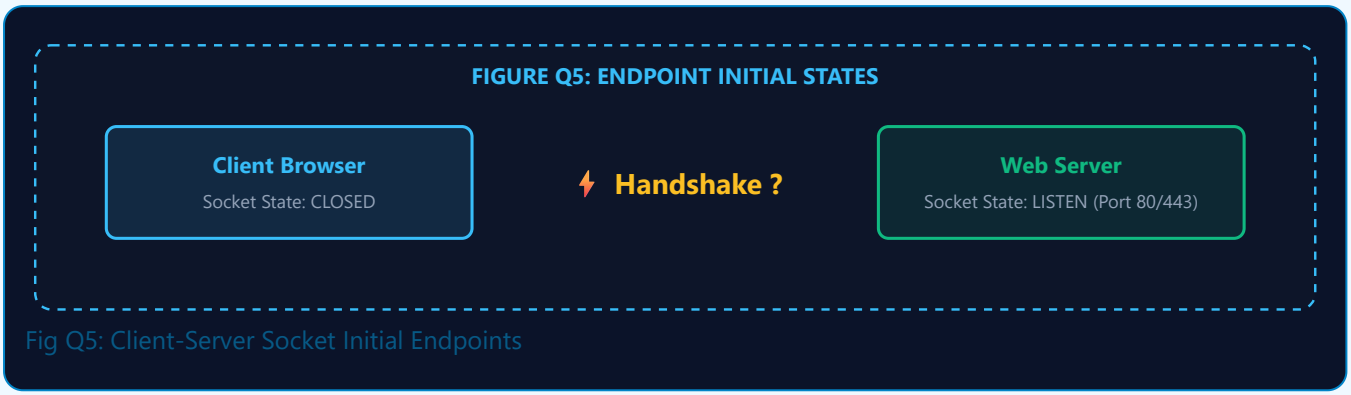


Fig Q5: Client-Server Socket Initial Endpoints

Step-by-Step Solution:

1. **SYN:** Client sends initial sequence number x with `SYN` flag set.

2. **SYN-ACK:** Server responds with `SYN` (ISN y) and `ACK = x + 1`.

3. **ACK:** Client confirms with `ACK = y + 1`. Connection established!

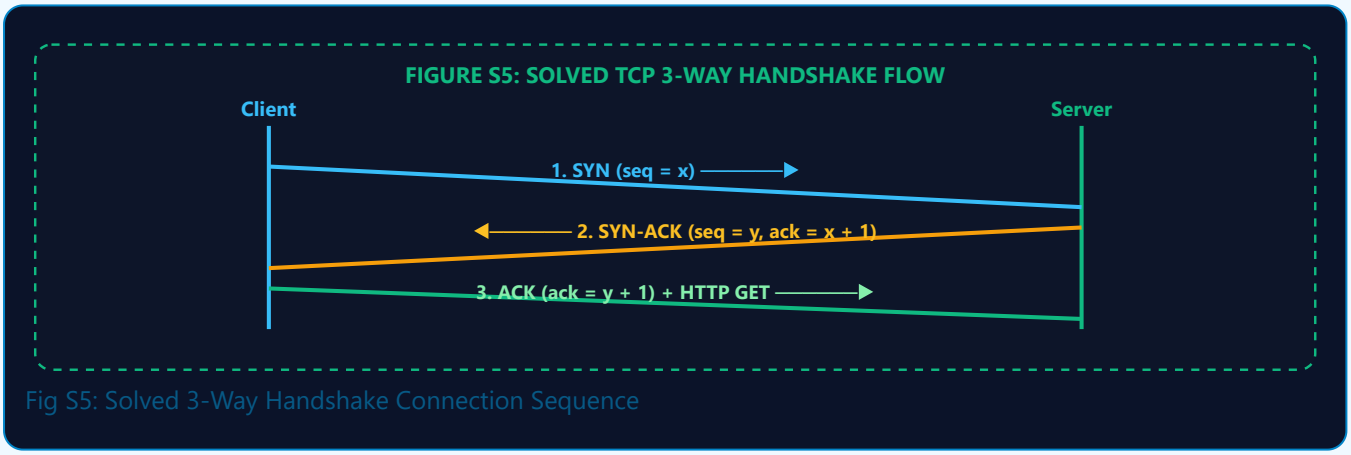


Fig S5: Solved 3-Way Handshake Connection Sequence

🎯 Handshake completes in 3 steps: SYN ➔ SYN-ACK ➔ ACK, transitioning sockets to ESTABLISHED state.